# 3W Well Data Viewer

## A desktop GUI to inspect the real instances of the 3W dataset

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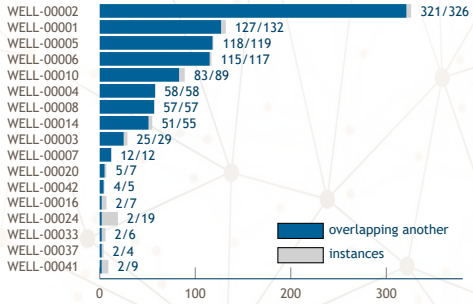
University of Brasília

5th Workshop on the 3W dataset
November 2026

# Overlapping samples

**Timelines page** ► How are the instances of one well distributed in time?

17(42.5%) wells have overlaps, totalling 986(88.11%) instances




Example: instances overlapping in WELL-00004.




Worries:

- Overlapping samples may land in **train and test**, causing leakage.
- Duplicated stretches pull **clusters** tighter, inflating scores.

Thankfully, 3W 2.0.0 has no label collisions, making joins principled and avoiding leakage.

Clustering must still be carefully calculated!

# What a joined history is labeled as

**Timelines page** ► How are the instances of one well distributed in time?

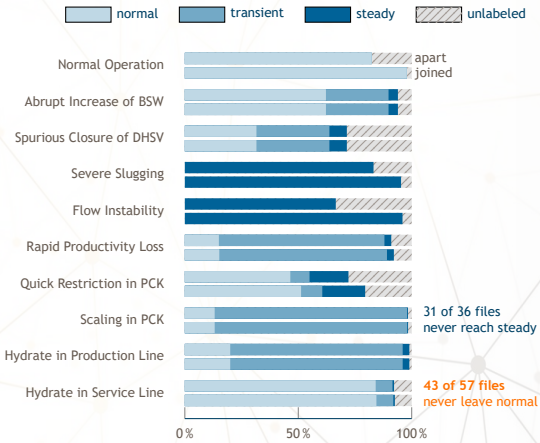
**12.3 % ▷ 2.9 %**
of unlabels after join

After the join, most unlabeled stretches appear trivial and might be recoverable.

| Unlabeled stretch position | *n* | Duration | Share |
|---|---|---|---|
| head of a recording | 208 | 1 h each | 86.5 % |
| seam between equal labels | 882 | median 76 s, max 56 min | 11.5 % |
| between two labels | 16 | median 6 min, max 1 h | 2.0 % |

▸ Appendix

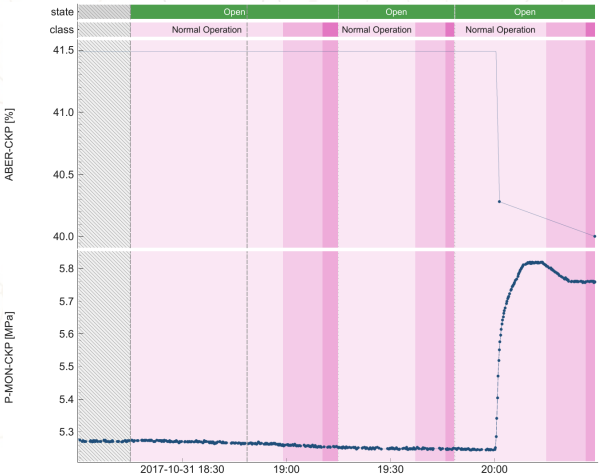
Each fault folder's recorded time, by label, its files apart and joined



# Histories whose labels do not add up

**Timelines page** ► How are the instances of one well distributed in time?

WELL-00004's joined Quick Restriction in PCK instances.



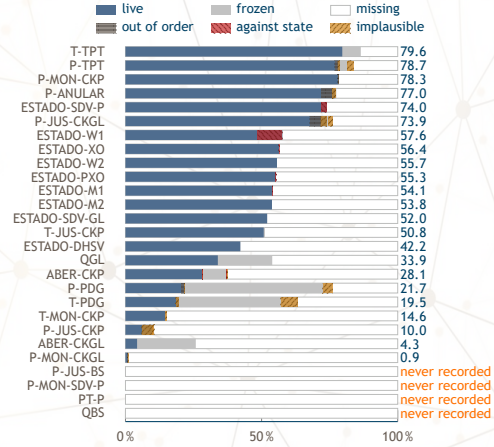

- Joined histories go back to normal **13 times** in 7 wells, **every time at a file boundary**.
- **WELL-00004**, left: three quick restrictions in 75 min, each back to normal 5 to 15 s after its steady state.
- **WELL-00006**: Abrupt Increase of BSW, steady, then normal 30 min later in a Scaling in PCK file.
- The others: Flow Instability (WELL-00001, 00004), Rapid Productivity Loss (00016, 00020), Scaling in PCK (00024), Hydrate in Service Line (00042).

Appendix

# What each variable recorded

**Availability page** ► How well do sensors and states cover the dataset?

See the detailed spread in the ▸ Appendix




**PL check** P-PDG → P-TPT → P-MON-CKP → *PCK open* → P-JUS-CKP

**SL check** P-MON-CKGL → *CKGL open* → P-JUS-CKGL ← *M2, W2 open* ← P-ANULAR

## Valve checks

| | |
|---|---|
| Open | M1, W1, SDV-P, PCK open; PXO, XO closed |
| Restart | M1, W1, SDV-P, PCK open |
| Shut-In, Closed With Diesel or Gas | one of M1, W1, SDV-P, PCK closed |
| Flushing Diesel or Gas | M1 or W1 closed; PXO or XO open |
| Depressurization | SDV-P, PCK open; M1 or W1 closed; PXO, XO closed |

## Plausible ranges

| Pressure | Temperature | Valve opening | Flow rate | Valve state |
|---|---|---|---|---|
| 0 to 100 MPa | −50 to 250 °C | 0 to 100 % | 0 to 10 $m^3/s$ | $<10^8$ |

# What each fault does to its signature

**Faults & Features pages** ► What does a fault do across sensors, and a sensor across faults?

| Fault | Inst. | Wells | From | Topside | | | Seabed | | Subsurface |
|---|---|---|---|---|---|---|---|---|---|
| | | | | ABER-CKP | P-MON-CKP | T-JUS-CKP | P-TPT | T-TPT | P-PDG |
| Normal Operation | 594 | 9 | | | | | | | |
| Abrupt Increase of BSW | 4 | 3 | thesis | | ↓3/3 (1) | ↑3/3 (1) | ↓4/4 | ↑4/4 | |
| Spurious Closure of DHSV | 22 | 7 | paper | | ↓6/7 (15) | | ↓21/21 (1) | ↓22/22 | ↑15/16 (6) |
| Severe Slugging | 32 | 2 | paper | | ∼ | ∼ | ∼27/32 | | ∼4/31 (1) |
| Flow Instability | 343 | 7 | thesis | | ∼(112) | ∼(195) | ∼(85) | | ∼(309) |
| Rapid Productivity Loss | 11 | 3 | thesis | | ↓11/11 | ↓10/11 | ↓6/10 (1) | ↓10/11 | |
| Quick Restriction in PCK | 6 | 2 | paper | ↓0/1 (5) | ↑1/3 (3) | | ↑4/6 | | ↑-(6) |
| Scaling in PCK | 36 | 6 | thesis | | ↑36/36 | ↓27/32 (4) | ↑29/31 (5) | ↓7/32 (4) | |
| Hydrate in Production Line | 14 | 9 | paper | | ↓9/14 | | ↑7/8 (5) | ↓6/7 (7) | ↑9/9 (5) |
| Hydrate in Service Line | 57 | 15 | none | | ? | | ?(32) | ?(27) | ?(28) |

↑ ↓ the move the 3W paper [1] or thesis [2] expects from the last hour of normal operation to the last hour of the event

∼ a swing

? no source

*a*/*b* instances that make it, of those that can show it (Severe Slugging: a swing over the paper's 10 bar).

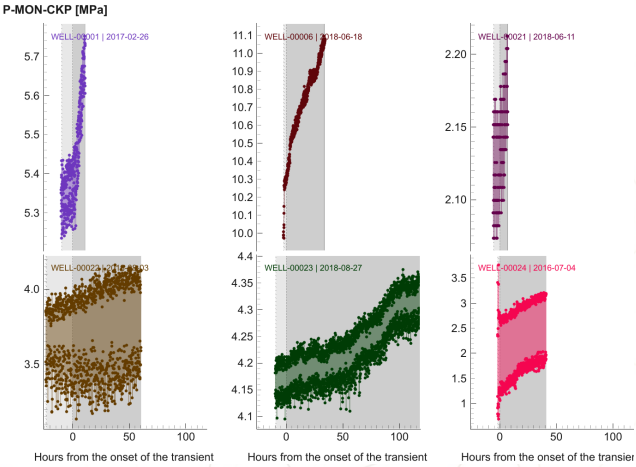
(m) instances where the variable is absent or frozen

Most signatures hold, but each has its exceptions, and some cannot be checked at all

# The same fault, well by well

**Faults & Features pages** ► What does a fault do across sensors, and a sensor across faults?

Scaling in PCK: P-MON-CKP from the onset of the transient, the earliest instance of each well

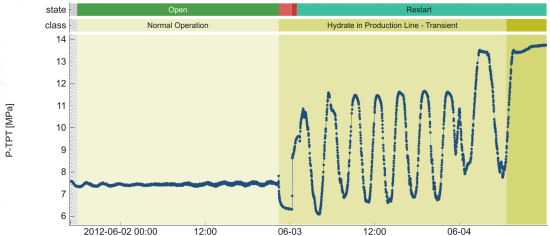



- **Scaling in PCK**: P-MON-CKP rises in 36 of 36, but T-TPT, expected to fall, rises in 15 of WELL-00024's 19 instances.
- **Rapid Productivity Loss**: WELL-00016's P-TPT rises in 4 of 4.
- **Quick Restriction in PCK**: P-PDG is frozen in 6 of 6 instances, the choke opening live in 1.
- **Severe Slugging**: swings P-TPT by 2.6 MPa (median), 27 of 32 over 10 bar, but P-PDG by 0.89 (4 of 31)
- **Flow Instability**: swings P-TPT by 0.07.

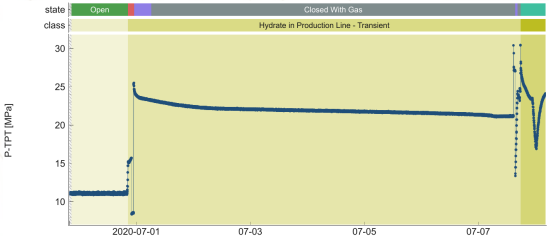
# Hydrate in Production Line: the ugly duckling

**Faults & Features pages** ► What does a fault do across sensors, and a sensor across faults?

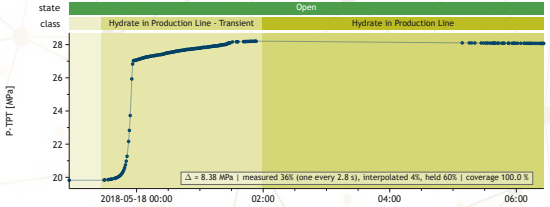
WELL-00019: steady means **plugged**.




WELL-00025: steady means **reopened**.




SIMULATED-00001: steady means **plugged**.



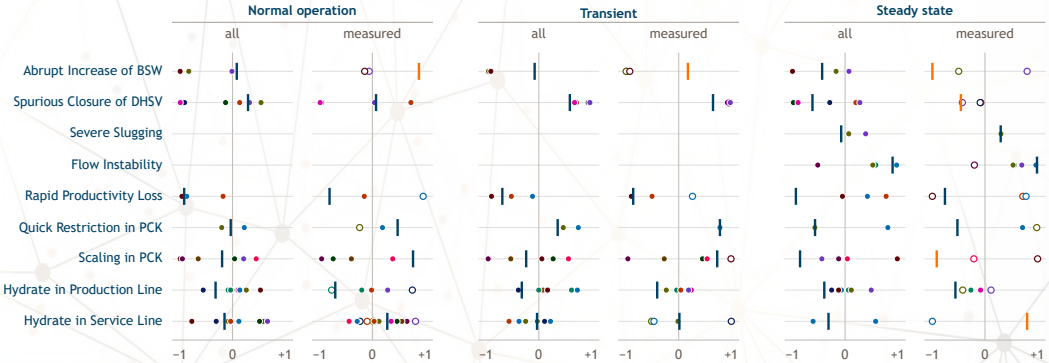

Hydrate labeling across instances, real or simulated, do not seem to share the same philosophy

Appendix

# Comparing wells

## Dispersions page ► How do sensors correlate across faults?

Each fault's pair ▸ Appendix: a dot per well in its color on the viewer's pages, hollow on fewer than 30 measurements; a bar for the fault's samples pooled, orange on fewer than 30 measurements

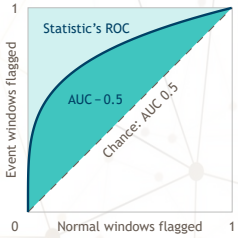



Pooled, a coefficient measures the gaps between wells.

# Separating normal from anomalous operation

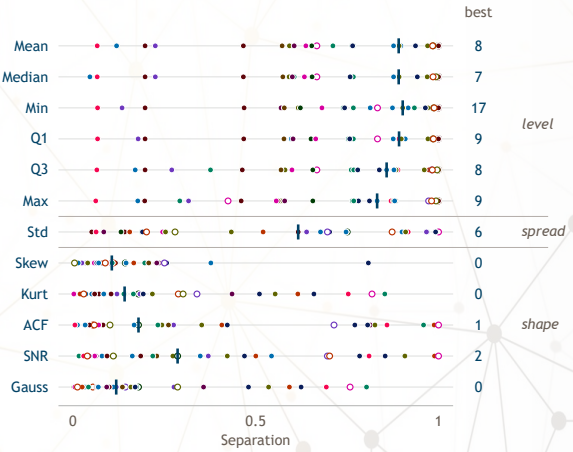
**Windows page** ► Can statistical features calculated on short horizons represent faults well?

- A binary classifier's separation power is AUC-based:




Separation better than chance
$= 2 \times |\mathrm{AUC} - 0.5|$
$=$ ▬ / (▬ + ▭)
$= 0.6$ here, of AUC 0.8

Statistics' separation ▸ Appendix on windows of 512 samples.




Short windows tell faults by their level, but may indicate when an instance was recorded rather than its fault.

# References


[1] R. E. V. Vargas, A. J. de Melo Júnior, C. B. de Campos Lima et al.
**3W Dataset 2.0.0: a realistic and public dataset with rare undesirable real events in oil wells.**
*Scientific Data* 13, 949 (2026).
doi:10.1038/s41597-026-07225-z

[2] R. E. V. Vargas.
**Base de dados e benchmarks para prognóstico de anomalias em sistemas de elevação de petróleo.**
Doctoral thesis, Universidade Federal do Espírito Santo, Vitória (2019).
http://repositorio.ufes.br/handle/10/13455

PETROBRAS

# Thank you!



github.com/Thiago-TP/3w-well-data-viewer

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# Appendix: label changes across an unlabeled stretch

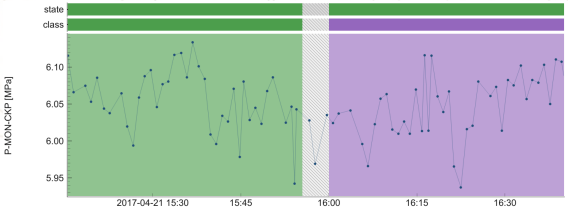
**Timelines page** ► How are the instances of one well distributed in time?

The 16 unlabeled stretches of a joined history between two different labels.

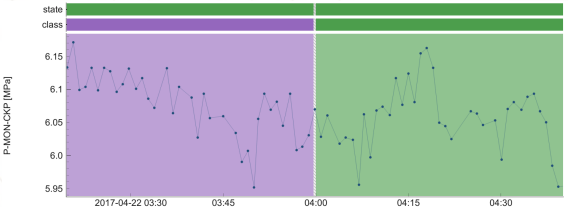
| | Well | Unlabeled from | Label before | Label after | For | Shown on |
|---|---|---|---|---|---|---|
| 1 | WELL-00001 | 2017-04-21 15:55 | normal | Flow Instability (steady) | 4 min | P-MON-CKP |
| 2 | WELL-00001 | 2017-04-22 03:59 | Flow Instability (steady) | normal | 25 s | P-MON-CKP |
| 3 | WELL-00001 | 2017-06-26 09:59 | Flow Instability (steady) | normal | 3 min | P-MON-CKP |
| 4 | WELL-00002 | 2013-12-12 08:59 | normal | Flow Instability (steady) | 41 s | P-TPT |
| 5 | WELL-00002 | 2013-12-14 22:00 | normal | Flow Instability (steady) | 9 s | P-ANULAR |
| 6 | WELL-00004 | 2014-08-06 22:00 | Flow Instability (steady) | normal | 32 s | P-MON-CKP |
| 7 | WELL-00004 | 2017-10-31 19:14 | Quick Restriction in PCK (steady) | normal | 5 s | P-MON-CKP |
| 8 | WELL-00004 | 2017-10-31 19:48 | Quick Restriction in PCK (steady) | normal | 15 s | P-MON-CKP |
| 9 | WELL-00006 | 2018-06-18 11:00 | Abrupt Increase of BSW (steady) | normal | 30 min | P-MON-CKP |
| 10 | WELL-00016 | 2018-04-26 14:44 | Rapid Productivity Loss (transient) | normal | 7 min | P-MON-CKP |
| 11 | WELL-00020 | 2014-03-18 01:30 | Rapid Productivity Loss (steady) | normal | 29 min | P-MON-CKP |
| 12 | WELL-00020 | 2014-03-18 13:57 | Rapid Productivity Loss (transient) | normal | 53 min | P-MON-CKP |
| 13 | WELL-00020 | 2014-03-19 11:00 | Rapid Productivity Loss (transient) | normal | 1 h | P-MON-CKP |
| 14 | WELL-00024 | 2016-09-11 07:00 | Scaling in PCK (transient) | normal | 40 min | P-MON-CKP |
| 15 | WELL-00042 | 2014-12-18 00:53 | Hydrate in Service Line (transient) | normal | 48 min | P-MON-CKP |
| 16 | WELL-00042 | 2014-12-22 03:37 | Hydrate in Service Line (transient) | normal | 8 min | P-MON-CKP |

# Appendix: label changes 1 to 4 (of 16)

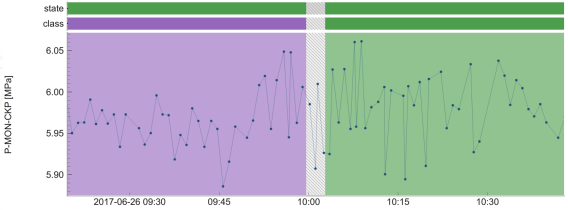
**Timelines page** ► How are the instances of one well distributed in time?



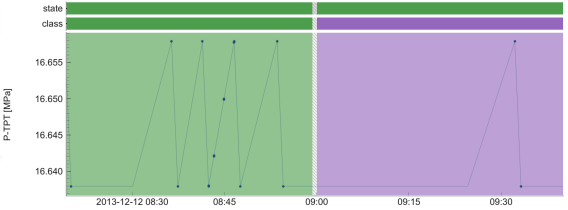

1 WELL-00001, 2017-04-21 15:55: normal → 4 min unlabeled → Flow Instability (steady)




2 WELL-00001, 2017-04-22 03:59: Flow Instability (steady) → 25 s unlabeled → normal



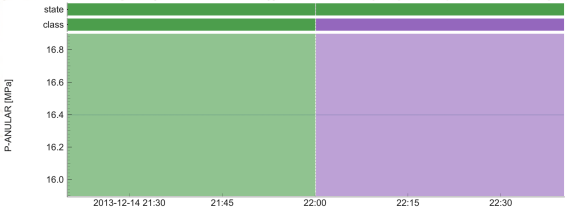

3 WELL-00001, 2017-06-26 09:59: Flow Instability (steady) → 3 min unlabeled → normal



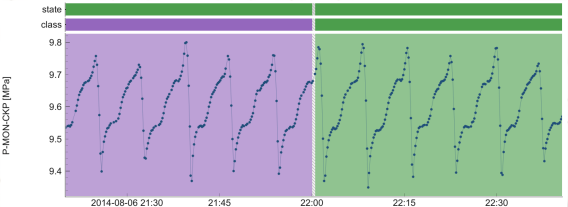

4 WELL-00002, 2013-12-12 08:59: normal → 41 s unlabeled → Flow Instability (steady)

# Appendix: label changes 5 to 8 (of 16)

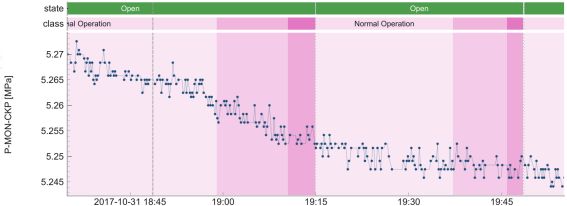
**Timelines page** ► How are the instances of one well distributed in time?



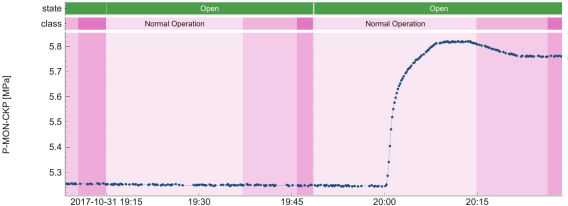

**5** WELL-00002, 2013-12-14 22:00: normal → 9 s unlabeled → Flow Instability (steady)




**6** WELL-00004, 2014-08-06 22:00: Flow Instability (steady) → 32 s unlabeled → normal



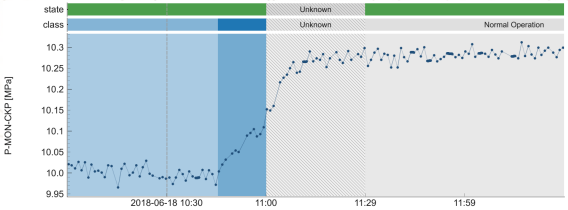

**7** WELL-00004, 2017-10-31 19:14: Quick Restriction in PCK (steady) → 5 s unlabeled → normal



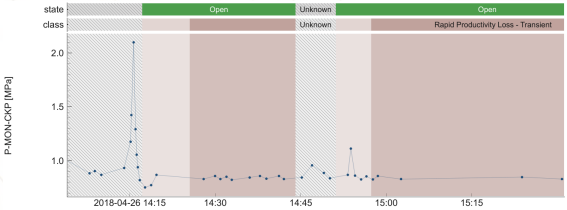

**8** WELL-00004, 2017-10-31 19:48: Quick Restriction in PCK (steady) → 15 s unlabeled → normal

# Appendix: label changes 9 to 12 (of 16)

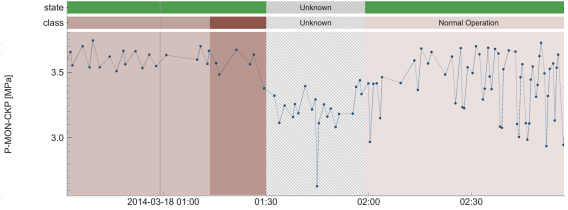
**Timelines page** ► How are the instances of one well distributed in time?



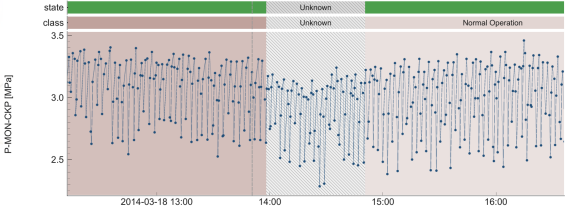

**9** WELL-00006, 2018-06-18 11:00: Abrupt Increase of BSW (steady) → 30 min unlabeled → normal




**10** WELL-00016, 2018-04-26 14:44: Rapid Productivity Loss (transient) → 7 min unlabeled → normal



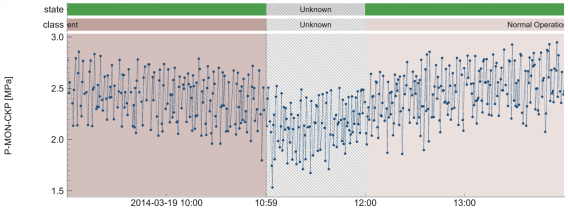

**11** WELL-00020, 2014-03-18 01:30: Rapid Productivity Loss (steady) → 29 min unlabeled → normal



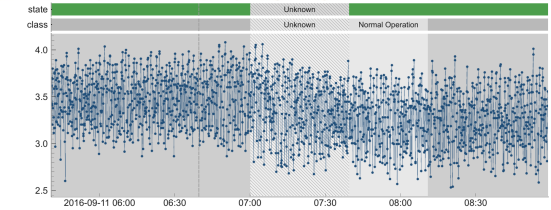

**12** WELL-00020, 2014-03-18 13:57: Rapid Productivity Loss (transient) → 53 min unlabeled → normal

# Appendix: label changes 13 to 16 (of 16)

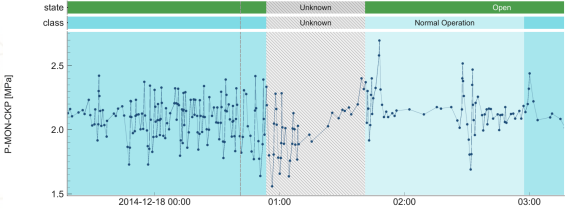
**Timelines page** ► How are the instances of one well distributed in time?



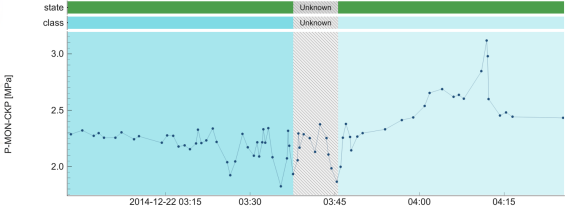

**13** WELL-00020, 2014-03-19 11:00: Rapid Productivity Loss (transient) → 1 h unlabeled → normal




**14** WELL-00024, 2016-09-11 07:00: Scaling in PCK (transient) → 40 min unlabeled → normal




**15** WELL-00042, 2014-12-18 00:53: Hydrate in Service Line (transient) → 48 min unlabeled → normal




**16** WELL-00042, 2014-12-22 03:37: Hydrate in Service Line (transient) → 8 min unlabeled → normal

# Appendix: what each variable recorded

**Availability page** ► How well do sensors and states cover the dataset?

| Variable | Live | | | Frozen | | | Implausible | | | Out of order | | | Against state | | | Missing | | | Farthest out |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | samples | % | inst. | samples | % | inst. | samples | % | inst. | samples | % | inst. | samples | % | inst. | samples | % | inst. | |
| T-TPT | 26,178,605 | 79.6 | 1,006 | 2,236,395 | 6.8 | 29 | 0 | 0.0 | 0 | - | - | - | - | - | - | 4,457,250 | 13.6 | 84 | |
| P-TPT | 25,878,401 | 78.7 | 922 | 1,612,066 | 4.9 | 38 | 1,004,849 | 3.1 | 10 | 361,705 | 1.1 | 29 | - | - | - | 5,381,783 | 16.4 | 159 | 2942 MPa |
| P-MON-CKP | 25,749,471 | 78.3 | 720 | 42,928 | 0.1 | 2 | 12,489 | 0.0 | 5 | 104,739 | 0.3 | 6 | - | - | - | 7,079,851 | 21.5 | 397 | −0.237 MPa |
| P-ANULAR | 25,320,751 | 77.0 | 914 | 139,764 | 0.4 | 7 | 387,650 | 1.2 | 3 | 1,296,207 | 3.9 | 6 | - | - | - | 7,411,735 | 22.5 | 198 | −7.658 MPa |
| ESTADO-SDV-P | 24,316,107 | 74.0 | 661 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | 653,430 | 2.0 | 23 | 8,556,143 | 26.0 | 458 | |
| P-JUS-CKGL | 24,307,254 | 73.9 | 656 | 743,125 | 2.3 | 33 | 1,299,132 | 4.0 | 85 | 1,298,303 | 4.0 | 7 | - | - | - | 7,821,871 | 23.8 | 430 | −0.4977 MPa |
| ESTADO-W1 | 18,938,036 | 57.6 | 558 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | 3,086,682 | 9.4 | 14 | 13,934,214 | 42.4 | 561 | |
| ESTADO-XO | 18,538,730 | 56.4 | 533 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | 3,351 | 0.0 | 1 | 14,333,520 | 43.6 | 586 | |
| ESTADO-W2 | 18,293,462 | 55.7 | 533 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 14,578,788 | 44.4 | 586 | |
| ESTADO-PXO | 18,176,203 | 55.3 | 532 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | 57,977 | 0.2 | 3 | 14,696,047 | 44.7 | 587 | |
| ESTADO-M1 | 17,773,205 | 54.1 | 501 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | 8,889 | 0.0 | 3 | 15,099,045 | 45.9 | 618 | |
| ESTADO-M2 | 17,688,282 | 53.8 | 504 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 15,183,968 | 46.2 | 615 | |
| ESTADO-SDV-GL | 17,105,271 | 52.0 | 480 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 15,766,979 | 48.0 | 639 | |
| T-JUS-CKP | 16,689,089 | 50.8 | 554 | 150,261 | 0.5 | 8 | 0 | 0.0 | 0 | - | - | - | - | - | - | 16,032,900 | 48.8 | 557 | |
| ESTADO-DHSV | 13,865,397 | 42.2 | 470 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 19,006,853 | 57.8 | 649 | |
| QGL | 11,150,903 | 33.9 | 147 | 6,569,202 | 20.0 | 298 | 0 | 0.0 | 0 | - | - | - | - | - | - | 15,152,145 | 46.1 | 674 | |
| ABER-CKP | 9,248,257 | 28.1 | 272 | 3,078,201 | 9.4 | 27 | 161,872 | 0.5 | 2 | - | - | - | 11,955 | 0.0 | 4 | 20,545,792 | 62.5 | 820 | −99.99 % |
| P-PDG | 7,119,658 | 21.7 | 174 | 17,836,376 | 54.3 | 653 | 1,200,682 | 3.7 | 33 | 256,966 | 0.8 | 24 | - | - | - | 7,916,216 | 24.1 | 292 | $-1.2\times10^{36}$ MPa |
| T-PDG | 6,426,251 | 19.5 | 142 | 14,385,021 | 43.8 | 421 | 2,445,015 | 7.4 | 36 | - | - | - | - | - | - | 12,060,978 | 36.7 | 556 | $-1.7\times10^{38}$ °C |
| T-MON-CKP | 4,789,008 | 14.6 | 14 | 161,538 | 0.5 | 1 | 161,538 | 0.5 | 1 | - | - | - | - | - | - | 27,921,704 | 84.9 | 1,104 | −99.99 °C |
| P-JUS-CKP | 3,287,363 | 10.0 | 10 | 161,538 | 0.5 | 1 | 1,506,941 | 4.6 | 7 | 0 | 0.0 | 0 | - | - | - | 29,423,349 | 89.5 | 1,108 | 1559 MPa |
| ABER-CKGL | 1,403,126 | 4.3 | 48 | 7,088,387 | 21.6 | 238 | 0 | 0.0 | 0 | - | - | - | - | - | - | 24,380,737 | 74.2 | 833 | |
| P-MON-CKGL | 308,726 | 0.9 | 2 | 0 | 0.0 | 0 | 6,401 | 0.0 | 1 | 2,096 | 0.0 | 1 | - | - | - | 32,563,524 | 99.1 | 1,117 | −0.094 MPa |
| P-JUS-BS | 0 | 0.0 | 0 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 32,872,250 | 100.0 | 1,119 | |
| P-MON-SDV-P | 0 | 0.0 | 0 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 32,872,250 | 100.0 | 1,119 | |
| PT-P | 0 | 0.0 | 0 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 32,872,250 | 100.0 | 1,119 | |
| QBS | 0 | 0.0 | 0 | 0 | 0.0 | 0 | 0 | 0.0 | 0 | - | - | - | - | - | - | 32,872,250 | 100.0 | 1,119 | |

# Appendix: hydrate instances transients and steadies

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**Faults & Features pages** ► What does a fault do across sensors, and a sensor across faults?

| Well | From | Transient | | | Steady | | |
|---|---|---|---|---|---|---|---|
| | | h | state | % | h | state | % |
| WELL-00019 | 2012-06-01 | 32 | Restart | 94 | 6 | Restart | 100 |
| WELL-00019 | 2014-11-17 | 73 | Shut-In | 98 | 2 | Flushing Diesel | 100 |
| WELL-00019 | 2015-06-14 | 30 | Shut-In | 92 | 1 | Bullheading | 100 |
| WELL-00019 | 2021-06-17 | 53 | Shut-In | 86 | 2 | Flushing Diesel | 100 |
| WELL-00025 | 2020-06-29 | 165 | Closed With Gas | 93 | 11 | Restart | 100 |
| WELL-00026 | 2016-06-01 | 43 | Open | 100 | 2 | Open | 100 |
| WELL-00026 | 2017-02-07 | 126 | Open | 100 | | never | |
| WELL-00026 | 2017-06-08 | 79 | Open | 100 | | never | |
| WELL-00027 | 2023-09-18 | 166 | Shut-In | 100 | 8 | Flushing Diesel | 100 |
| WELL-00028 | 2021-06-17 | 76 | Shut-In | 65 | 2 | Restart | 100 |
| WELL-00029 | 2020-10-20 | 23 | Shut-In | 72 | 1 | Bullheading | 100 |
| WELL-00030 | 2014-11-08 | 33 | Shut-In | 81 | 2 | Flushing Diesel | 100 |
| WELL-00031 | 2014-04-03 | 92 | Shut-In | 90 | 1 | Bullheading | 100 |
| WELL-00032 | 2011-08-30 | 24 | Shut-In | 98 | 2 | Flushing Diesel | 100 |
| 81 simulated | | | Open | 100 | | Open | 100 |

# Appendix: every fault's coefficients

**Dispersions page** ► How do sensors correlate across faults?

| Fault | Pair | Normal operation | | | Transient | | | Steady state | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | all | measured | | all | measured | | all | measured | |
| Abrupt Increase of BSW | P-TPT × T-TPT | +0.08 | +0.88 | (24) | −0.08 | +0.17 | (20) | −0.44 | −1.00 | (6) |
| Spurious Closure of DHSV | P-TPT† × T-TPT | +0.29 | +0.07 | (314) | +0.59 | +0.65 | (494) | −0.62 | −0.46 | (28) |
| Severe Slugging | P-TPT × T-JUS-CKP | - | - | | - | - | | −0.07 | +0.30 | (1,348) |
| Flow Instability | P-MON-CKP† × P-TPT† | - | - | | - | - | | +0.90 | +0.99 | (11,999) |
| Rapid Productivity Loss | P-ANULAR × T-TPT | −0.92 | −0.82 | (366) | −0.69 | −0.86 | (394) | −0.94 | −0.76 | (42) |
| Quick Restriction in PCK | P-ANULAR × P-TPT | −0.03 | +0.48 | (1,132) | +0.36 | +0.78 | (420) | −0.57 | −0.52 | (130) |
| Scaling in PCK | P-JUS-CKGL × P-MON-CKP | −0.20 | +0.77 | (1,555) | −0.24 | +0.73 | (12,891) | −0.86 | −0.92 | (16) |
| Hydrate in Production Line | P-MON-CKP × QGL | −0.32 | −0.71 | (7,349) | −0.32 | −0.41 | (22,644) | −0.40 | −0.56 | (493) |
| Hydrate in Service Line | P-JUS-CKGL × P-MON-CKP | −0.15 | +0.28 | (14,053) | −0.03 | +0.01 | (3,125) | −0.32 | +0.80 | (15) |

Pair of best covering variables available in all instances of a fault

† instances of a fault don't share a pair and defaulted to the best covered of its signature [1, 2]

all Pearson's coefficient over every plausible sample of the period, pooled over the fault's instances

measured over the samples at which both were measured, their count in brackets, set apart below 30

# Appendix: every well's coefficients (1 of 3)

PETROBRAS

**Dispersions page** ► How do sensors correlate across faults?

| Fault | Pair | Well | Normal operation | | Transient | | Steady state | |
|---|---|---|---|---|---|---|---|---|
| | | | all | measured | all | measured | all | measured |
| Abrupt Increase of BSW | P-TPT × T-TPT | WELL-00001 | −0.01 | −0.06 (5) | −0.97 | −0.99 (6) | +0.07 | +0.80 (3) |
| | | WELL-00002 | −0.83 | - | −0.96 | −1.00 (4) | −0.17 | −0.50 (3) |
| | | WELL-00006 | −0.99 | −0.15 (18) | −0.91 | −0.93 (10) | −1.00 | - |
| Spurious Closure of DHSV | P-TPT† × T-TPT | WELL-00002 | +0.54 | - | +0.97 | +0.96 (6) | - | - |
| | | WELL-00003 | +0.14 | +0.73 (95) | +0.72 | +0.93 (51) | +0.20 | −0.08 (3) |
| | | WELL-00009 | −0.13 | - | +0.95 | +0.94 (11) | −0.98 | - |
| | | WELL-00011 | −0.91 | −0.96 (115) | +0.94 | +0.92 (287) | −0.29 | −0.09 (17) |
| | | WELL-00012 | −0.99 | −0.99 (53) | +0.68 | +0.97 (4) | −0.89 | - |
| | | WELL-00013 | +0.32 | +0.04 (51) | +0.97 | +0.98 (135) | +0.28 | −0.43 (5) |
| Severe Slugging | P-TPT × T-JUS-CKP | WELL-00001 | - | - | - | - | +0.39 | - |
| | | WELL-00014 | - | - | - | - | +0.07 | +0.30 (1,347) |
| Flow Instability | P-MON-CKP† × P-TPT† | WELL-00001 | - | - | - | - | +0.57 | +0.70 (367) |
| | | WELL-00004 | - | - | - | - | +0.98 | +0.97 (126) |
| | | WELL-00005 | - | - | - | - | +0.59 | +0.56 (6,253) |
| | | WELL-00007 | - | - | - | - | −0.52 | −0.20 (12) |
| | | WELL-00014 | - | - | - | - | +0.53 | +0.53 (5,241) |
| Rapid Productivity Loss | P-ANULAR × T-TPT | WELL-00015 | −0.18 | −0.15 (360) | −0.52 | −0.51 (272) | +0.78 | +0.72 (25) |
| | | WELL-00016 | −0.87 | +0.96 (5) | −0.12 | +0.26 (4) | +0.42 | +0.78 (15) |
| | | WELL-00020 | −0.96 | - | −0.90 | −0.90 (118) | −0.05 | −1.00 (2) |

# Appendix: every well's coefficients (2 of 3)

**Dispersions page** ► How do sensors correlate across faults?

| Fault | Pair | Well | Normal operation | | Transient | | Steady state | |
|---|---|---|---|---|---|---|---|---|
| | | | all | measured | all | measured | all | measured |
| Quick Restriction in PCK | P-ANULAR × P-TPT | WELL-00002 | −0.20 | −0.24 (6) | +0.46 | - | −0.57 | +0.98 (7) |
| | | WELL-00004 | +0.22 | +0.19 (1,126) | +0.75 | +0.78 (420) | +0.81 | +0.71 (123) |
| Scaling in PCK | P-JUS-CKGL × P-MON-CKP | WELL-00001 | +0.21 | - | −0.95 | - | −0.44 | - |
| | | WELL-00006 | −1.00 | - | +0.06 | +0.99 (4) | +0.99 | +1.00 (2) |
| | | WELL-00021 | −0.95 | −0.96 (89) | −0.96 | −0.97 (77) | −0.12 | - |
| | | WELL-00022 | −0.65 | −0.40 (256) | −0.53 | −0.27 (1,769) | - | - |
| | | WELL-00023 | +0.04 | −0.75 (111) | +0.27 | +0.45 (1,245) | - | - |
| | | WELL-00024 | +0.45 | +0.39 (1,099) | +0.56 | +0.54 (9,796) | +0.04 | −0.21 (13) |
| Hydrate in Production Line | P-MON-CKP × QGL | WELL-00019 | −0.08 | −0.20 (1,687) | +0.62 | −0.04 (1,524) | +0.06 | −0.27 (299) |
| | | WELL-00025 | −0.56 | +0.76 (18) | −0.39 | - | −0.26 | - |
| | | WELL-00026 | +0.10 | +0.28 (4,869) | +0.11 | +0.24 (19,618) | −0.10 | −0.08 (183) |
| | | WELL-00027 | +0.25 | +0.29 (113) | +0.18 | +0.19 (216) | +0.50 | +0.12 (5) |
| | | WELL-00028 | +0.26 | - | +0.12 | −0.24 (1,176) | +0.12 | −0.42 (5) |
| | | WELL-00029 | −0.01 | −0.02 (654) | +0.08 | +0.04 (108) | - | - |
| | | WELL-00030 | +0.14 | - | +0.73 | - | −0.07 | - |
| | | WELL-00031 | −0.04 | −0.77 (8) | −0.00 | - | - | - |
| | | WELL-00032 | +0.53 | - | +0.17 | - | −0.12 | - |

# Appendix: every well's coefficients (3 of 3)

**Dispersions page** ► How do sensors correlate across faults?

| Fault | Pair | Well | Normal operation | | Transient | | Steady state | |
|---|---|---|---|---|---|---|---|---|
| | | | all | measured | all | measured | all | measured |
| Hydrate in Service Line | P-JUS-CKGL × P-MON-CKP | WELL-00010 | −0.28 | −0.30 (111) | - | - | - | - |
| | | WELL-00014 | +0.14 | +0.13 (2,330) | −0.03 | −0.01 (3,085) | - | - |
| | | WELL-00015 | −0.03 | −0.10 (6) | - | - | - | - |
| | | WELL-00016 | +0.59 | +0.82 (33) | +0.22 | - | +0.58 | - |
| | | WELL-00020 | −0.77 | - | - | - | - | - |
| | | WELL-00033 | +0.66 | +0.66 (2,285) | - | - | - | - |
| | | WELL-00034 | +0.58 | +0.56 (2,518) | - | - | - | - |
| | | WELL-00035 | +0.51 | +0.47 (3,998) | - | - | - | - |
| | | WELL-00036 | −0.30 | −0.44 (1,560) | - | - | - | - |
| | | WELL-00037 | −0.31 | −0.24 (21) | +0.11 | +1.00 (2) | - | - |
| | | WELL-00038 | −0.02 | +0.36 (37) | - | - | - | - |
| | | WELL-00039 | +0.67 | +0.82 (10) | - | - | - | - |
| | | WELL-00040 | −0.09 | +0.02 (20) | −0.25 | −0.52 (8) | - | - |
| | | WELL-00041 | −0.03 | +0.03 (1,092) | −0.57 | - | - | - |
| | | WELL-00042 | +0.12 | −0.29 (32) | −0.38 | −0.47 (28) | −0.61 | −1.00 (2) |

# Appendix: every statistic, well by well (1 of 2)

PETROBRAS

**Windows page** ► Can statistical features calculated on short horizons represent faults well?

Each well's best covered analog sensor (most samples live; ties, Tied, to the one separating best). A well's best separation at 512 samples in bold, its line orange on fewer than 30 windows of a kind; across window lengths it moves by 0.07 at most.

| Well | Sensor | Tied | Windows | | Level | | | | | | Spread | Shape | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | normal | event | Mean | Median | Min | Q1 | Q3 | Max | Std | Skew | Kurt | ACF | SNR | Gauss |
| WELL-00001 | P-TPT | 4 | 3,454 | 680 | 0.23 | 0.23 | 0.13 | 0.18 | 0.27 | 0.32 | **0.64** | 0.03 | 0.12 | 0.28 | 0.37 | 0.12 |
| WELL-00002 | T-TPT | 1 | 7,291 | 1,683 | 0.59 | 0.59 | **0.62** | 0.60 | 0.58 | 0.57 | 0.43 | 0.02 | 0.01 | 0.24 | 0.27 | 0.00 |
| WELL-00003 | P-MON-CKP | 1 | 949 | 40 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.97 | 0.52 | 0.23 | 0.62 | 0.99 | **0.99** | 0.59 |
| WELL-00004 | P-MON-CKP | 1 | 181 | 629 | 0.12 | 0.05 | 0.81 | 0.60 | 0.17 | 0.18 | **0.90** | 0.38 | 0.06 | 0.00 | 0.08 | 0.11 |
| WELL-00005 | P-TPT | 4 | 1,942 | 542 | 0.71 | 0.60 | 0.93 | 0.88 | 0.38 | 0.29 | **0.97** | 0.20 | 0.85 | 0.96 | 0.91 | 0.80 |
| WELL-00006 | T-TPT | 1 | 4,243 | 541 | 0.20 | 0.20 | 0.20 | **0.20** | 0.20 | 0.20 | 0.05 | 0.12 | 0.10 | 0.08 | 0.09 | 0.10 |
| WELL-00007 | P-JUS-CKGL | 6 | 30 | 147 | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | 0.92 | 0.23 | 0.44 | 0.81 | 0.78 | 0.36 |
| WELL-00008 | P-JUS-CKGL | 1 | 1,994 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00009 | P-ANULAR | 6 | 7 | 8 | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | 0.75 | 0.14 | 0.18 | 0.18 | 0.29 | 0.18 |
| WELL-00010 | T-TPT | 1 | 419 | 1,192 | 0.89 | 0.89 | 0.90 | 0.89 | 0.89 | 0.87 | **0.92** | 0.25 | 0.75 | 0.86 | 0.81 | 0.69 |
| WELL-00011 | P-TPT | 2 | 91 | 140 | 0.94 | 0.94 | **0.99** | 0.97 | 0.84 | 0.78 | 0.70 | 0.26 | 0.51 | 0.77 | 0.78 | 0.48 |
| WELL-00012 | P-ANULAR | 7 | 22 | 6 | 0.67 | 0.67 | 0.83 | 0.83 | 0.67 | 0.42 | **1.00** | 0.11 | 0.82 | **1.00** | **1.00** | 0.76 |
| WELL-00013 | P-ANULAR | 3 | 8 | 14 | **1.00** | **1.00** | **1.00** | **1.00** | 0.98 | 0.97 | 0.70 | 0.25 | 0.34 | 0.71 | 0.70 | 0.14 |
| WELL-00014 | QGL | 1 | 99 | 1,530 | 0.97 | 0.97 | **0.97** | 0.97 | 0.96 | 0.88 | 0.91 | 0.11 | 0.55 | 0.83 | 0.91 | 0.53 |
| WELL-00015 | ABER-CKP | 8 | 41 | 20 | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | 0.87 | 0.11 | 0.29 | 0.05 | 0.70 | 0.05 |
| WELL-00016 | T-TPT | 9 | 120 | 56 | 0.90 | 0.89 | **0.92** | 0.91 | 0.86 | 0.83 | 0.20 | 0.15 | 0.04 | 0.01 | 0.16 | 0.03 |
| WELL-00019 | QGL | 2 | 757 | 1,407 | 0.76 | 0.76 | 0.74 | 0.75 | 0.76 | **0.76** | 0.75 | 0.11 | 0.17 | 0.05 | 0.02 | 0.17 |
| WELL-00020 | T-TPT | 2 | 110 | 603 | 0.47 | 0.47 | **0.47** | 0.47 | 0.46 | 0.46 | 0.14 | 0.08 | 0.07 | 0.08 | 0.23 | 0.06 |
| WELL-00021 | P-MON-CKP | 6 | 88 | 87 | **0.60** | 0.60 | 0.57 | 0.60 | 0.60 | 0.58 | 0.08 | 0.23 | 0.08 | 0.07 | 0.06 | 0.09 |

# Appendix: every statistic, well by well (2 of 2)

PETROBRAS

**Windows page** ► Can statistical features calculated on short horizons represent faults well?

| Well | Sensor | Tied | Windows | | Level | | | | | | Spread | Shape | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | normal | event | Mean | Median | Min | Q1 | Q3 | Max | Std | Skew | Kurt | ACF | SNR | Gauss |
| WELL-00022 | T-JUS-CKP | 1 | 844 | 5,169 | 0.57 | 0.57 | **0.58** | 0.58 | 0.57 | 0.56 | 0.20 | 0.21 | 0.14 | 0.26 | 0.23 | 0.13 |
| WELL-00023 | P-MON-CKP | 1 | 361 | 2,664 | 0.65 | 0.65 | 0.62 | 0.65 | **0.66** | 0.66 | 0.13 | 0.01 | 0.02 | 0.10 | 0.11 | 0.02 |
| WELL-00024 | P-JUS-CKGL | 1 | 644 | 4,675 | 0.07 | 0.07 | **0.07** | 0.07 | 0.07 | 0.06 | 0.06 | 0.01 | 0.00 | 0.00 | 0.03 | 0.01 |
| WELL-00025 | P-JUS-CKP | 2 | 168 | 1,237 | 0.99 | 0.99 | 0.96 | 0.99 | 0.99 | 0.99 | **0.99** | 0.81 | 0.66 | 0.81 | 0.85 | 0.62 |
| WELL-00026 | T-MON-CKP | 8 | 754 | 1,758 | 0.64 | 0.63 | **0.68** | 0.66 | 0.60 | 0.56 | 0.24 | 0.06 | 0.00 | 0.07 | 0.06 | 0.01 |
| WELL-00027 | P-MON-CKP | 5 | 96 | 1,227 | 1.00 | 1.00 | **1.00** | **1.00** | 1.00 | 1.00 | 0.97 | 0.02 | 0.18 | 0.04 | 0.04 | 0.28 |
| WELL-00028 | T-MON-CKP | 1 | 86 | 545 | 0.89 | 0.89 | **0.90** | 0.90 | 0.88 | 0.88 | 0.25 | 0.04 | 0.22 | 0.35 | 0.42 | 0.16 |
| WELL-00029 | P-JUS-CKP | 8 | 112 | 169 | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | 0.99 | 0.15 | 0.17 | 0.02 | 0.41 | 0.50 | 0.03 |
| WELL-00030 | ABER-CKGL | 10 | 62 | 248 | **1.00** | **1.00** | 1.00 | **1.00** | **1.00** | **1.00** | 0.75 | 0.01 | 0.05 | 0.17 | 0.35 | 0.01 |
| WELL-00031 | T-MON-CKP | 2 | 51 | 651 | 0.77 | 0.77 | 0.77 | 0.77 | 0.77 | **0.77** | 0.19 | 0.07 | 0.14 | 0.23 | 0.22 | 0.04 |
| WELL-00032 | P-TPT | 5 | 71 | 185 | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | **1.00** | 0.62 | 0.00 | 0.06 | 0.04 | 0.19 | 0.07 |
| WELL-00033 | P-JUS-CKGL | 3 | 291 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00034 | P-JUS-CKGL | 3 | 365 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00035 | P-JUS-CKGL | 3 | 510 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00036 | P-JUS-CKGL | 4 | 35 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00037 | P-JUS-CKGL | 7 | 129 | 30 | 0.77 | 0.76 | 0.74 | 0.76 | 0.78 | **0.81** | 0.19 | 0.07 | 0.19 | 0.42 | 0.47 | 0.17 |
| WELL-00038 | ABER-CKP | 9 | 236 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00039 | P-ANULAR | 5 | 116 | 0 | - | - | - | - | - | - | - | - | - | - | - | - |
| WELL-00040 | P-ANULAR | 5 | 35 | 26 | 0.98 | 0.99 | 0.99 | 0.98 | **1.00** | 0.99 | 0.28 | 0.00 | 0.30 | 0.10 | 0.11 | 0.29 |
| WELL-00041 | P-PDG | 2 | 1,893 | 23 | 0.98 | 0.98 | **0.99** | 0.99 | 0.98 | 0.98 | 0.20 | 0.09 | 0.03 | 0.06 | 0.04 | 0.01 |
| WELL-00042 | P-JUS-CKGL | 9 | 99 | 190 | 0.88 | 0.88 | 0.87 | 0.88 | 0.88 | **0.88** | 0.68 | 0.07 | 0.08 | 0.03 | 0.54 | 0.07 |